# iLLD-TC4-v2.5.0

## Release Notes

Product name: TC4xx_SW_iLLD
Release number and version: iLLD-TC4-v2.5.0
Type of release: Alpha*
Release method: Github
Processor platform: TC4Dx AA, TC4Dx AB
Date: 2026-02-27
Previous release number and version: iLLD-TC4-v2.4.1

## About this document

### Scope and purpose

This document details the release contents, all known issues in the release, and the changes from the last release, together with information on tools and support changes.

New issues identified since the last release of this document are detailed first, followed by all issues identified in previous versions of this release.

*Note: * This release is not intended for production use. This is reference software that helps the customers to develop their software.*

***Attention: Refer to the Limitations and deviations chapter before using the software for integration.***

### Intended audience

This document is intended for anyone using the iLLD software.

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# 1 Release contents

## 1.1 Release overview

This is Alpha Release of iLLD SW for the AURIX™ 3G TC4xx platform.

## 1.2 Released items

This release contains the release items mentioned in Table 1

**Table 1 Release content**

| Package content | Description |
|---|---|
| iLLD-TC4-v2.5.0.zip | iLLD package |

iLLD-TC4-v2.5.0.zip contains the following items mentioned in Table 2

**Table 2 Release zip content**

| Folder/File | Description |
|---|---|
| doc | iLLD documentation |
| examples | iLLD template projects |
| src | iLLD source for TC4Dx device |
| iLLD-TC4-v2.5.0-Releasenotes.pdf | A3G iLLD Release Notes |

The Hardware functional blocks supported in TC4Dx are mentioned in Table 3

**Table 3**

| Hardware functional block | iLLD driver |
|---|---|
| CPU | Cpu, Stm |
| NVM | Flash |
| SDMA | Dma |
| PPUC | Ppuc |
| GETH | Geth |
| HSPHY | Hsphy |
| HSSL, HSCT | Hssl |
| XSPI | Xspi |
| PCIE | Ep, Rc |
| DRE | Dre |
| MCMCAN | Can, Cre |
| ERAY | Eray |
| CAN-XL | Canxl |
| ASCLIN | Asc, Spi, Lin |

| | |
|---|---|
| SENT | Sent |
| PSI5 | Psi5 |
| PSI5s | Psi5s |
| MSC | Msc |
| SDMMC | Sd, Emmc |
| QSPI | QspiMaster, QspiSlave |
| I2C | I2c |
| CLOCK | Clock |
| VMT | Vmt |
| SCU | Eru |
| PMS | Pms |
| SMM | SmmReset, SmmSysMode |
| IR | Src |
| WTU | Wtu |
| SMU | Smu, SmuStdby |
| FCE | Fce |
| eGTM | Tom, Atom, Tim, Trigger, Pwm, Input |
| ADC | Tmadc, Cdsp, Dsadc, Fcc |
| LETH | Leth |
| PORTs | Port |
| AP | Apu, Prot |
| SCR | AdcComp, Cpu, Port, Rtc, Scu, Ssc, T2ccu, Timers, Uart, Wcan,I2C |

## 1.3 Compatibility

The iLLD software now includes support for C++ compilation. To accommodate this new feature, several modifications have been made to the make options. For users who are utilizing iLLD in C projects and do not wish to alter their make options, it is necessary to incorporate specific C++ dummy functions into the "Ifx_Cfg_Ssw.c" file to enable successful compilation.

```
#ifdef __TASKING__
void _main(void)
{}
void _doexit(void)
{}
#elif defined(__HIGHTEC__) && !defined(__clang__)
void _init(void)
{}
#elif defined(__GNUC__) && !defined(__HIGHTEC__)
void _init(void)
{}
#elif defined(__ghs__)
void _main(void)
{}
void exit(int a)
{}
#endif
```

Add "atexit" macro in Ifx_Cfg_Ssw.h
#define atexit Ifx_atexit

Add below lines of code in Ifx_Cfg_Ssw.c file:

```
#if defined(HIGHTEC) && defined(clang)
void __libc_init_array(void)
{}

void __libc_fini_array (void)
{}

int Ifx_atexit(void (*func)(void))
{
return 0;
}

#endif
```

# 2 Tool information

The tool information is provided in the following table.

**Table 4 Tool details**

| Tool description | Version details |
|---|---|
| Compiler | Hightec LLVM v10.1.0<br>TC CTC SmartCode v10.4r1<br>Metaware_Aurix v2.1/ TC CARC SmartCode v10.4r1<br>GreenHills v2025.1<br>Windriver v5.9.9.0<br>GCC 11.3.1 20221230 Built 2025-09-08 |
| Processor platform | TC4Dx |
| Evaluation hardware | TC4Dx_VDK_v2.3.0<br>TC4Dx: Target hardware (Triboard)<br>TC4Dx: Target hardware (Emulation) |

# 3 Summary of changes

## 3.1 Issues fixed in release V2.5.0

| Issue type | Summary | Description |
|---|---|---|
| Bug | TC4Dx PPU Clock config wrong | [8896] Added support for K3PREDIV and K3DIV support in SYSPLLCON1 register to derive the ppupll clock. |
| Bug | Possible bug in IfxVmt.c TC4xx iLLD 2.4.0-EIR | [9070] Removed unwanted masking which writes to reserved bits when accessing MCONTROL register in IfxVmt_readSramAddress and IfxVmt_writeSramAddress API |
| Bug | [TC4xx][CDSP]: Using GCC compiler with O3 optimizations enabled leads to a trap when clearing CDSP memories | [9112] Added Volatile keyword to prevent the optimization |
| Bug | EGTM TIM confusing enum elements | [9241] Enums are renamed to better reflect its function from a signal point of view |
| Bug | I2C: No handling for FIFO Underflow Errors in I2c_read and I2c_write APIs | [9255] FIFO Underflow handling support added for read and write APIs |
| Bug | Instruction trap caused by HRA_CORE_ID reading without mfcr | [9271] Updated IfxStm_initCompare Api to get index from passed cpu parameter using different way rather than using HRA_CORE_ID which used to cause trap |
| Bug | TC4x HSPHY XPCS: IfxHsphy_Xpcs_setSgmiiSpeedMode seqeunce different compared to UM | [9279] Initialization sequence updated for XPCS |
| Bug | xSPI: Fix IfxXspi_Status enum | [9346] Removed IfxXspi_Status_noError as its redundant |
| Bug | IfxCpu_Trap_extractTrapInfo not returning the expected values in each core while using GCC | [8519]IfxCpu_Trap_extractTrapInfo not returning the expected values in each core while using GCC compiler |
| Bug | [TC4xx][CDSP]: Wrong usage of CDSP result register RES0 in IfxAdc_Cdsp_readDspCoreResult | [8753] Returned the valid result from register if event flag is not set |
| Bug | IfxCanxl_Can_stopNode is not using the correct LOCK register | [8899] Register MH_LOCK used for unlock sequence over Ni_LOCK |
| Bug | xSPI: Wrong ARLEN and AWLEN configuration in IfxXspi_Spi_dmaEnhancedDataExchange | [8906] The configuration of ARLEN and AWLEN burst lengths for internal DMA is updated to use correct TMOD |
| Bug | CPU Generic File includes Tricore specific file | [8993]XSPI driver is moved to cpugeneric from tricore, due to dependency for hsphy which is present in cpugeneric |

| | | |
|---|---|---|
| Bug | bug for get the frequencies | [9007] Fixed the rounding off issue in IfxClock_getPllFrequency, IfxClock_getPerPllFrequency1, IfxClock_getPerPllFrequency2, IfxClock_getPerPllFrequency3 APIs |
| Bug | Linker script of TASKING for PPU needs correction. | [9056] The linker script is updated for PPU for defining the stack size |
| Bug | CompilerHightec.c creates compilation issue in PPU | [9064] Conditional check if added to include the ssw file only when compiling for tricore and avoid when building ppu |
| Bug | ASClIN refers to Services files | [9065] Dependency with service layer removed for ASCLIN |
| Bug | Not all system services are compilable for PPU | [9066] The SysSe folder is located in the CPUGeneric folder but COMM and MATH libraries inside it are not compilable for PPU either they need to be moved out of CPUGeneric or made compilable for all ISAs |
| Bug | IfxCan_Can_readTxEvent incorrect timestamp reading | [9068] Timestamp read logic updated to use E1B register |
| Bug | IfxCan_Can_initNode() does not initailize timeStampType of IfxCan_Can_Node | [9069] Configuration for timestampType is updated in module handle for correct reference during initialization |
| Bug | Ambiguity and Potential Misuse of IfxCan_FilterType_none in Extended Filter Configuration | [9257] Documentation updated with correct details about Basic and Extended filter types |
| Enhancement | EGTM PWM driver interrupts cannot be configured externally | [4549] Updated PWM interrupt configuration to check for priority instead of callback pointers. This change provides more flexibility and ensures consistency across the Egtm modules. |
| Enhancement | TC4x LETH: General update for Optimizing APIs in IfxLeth_Eth.c | [8328] Ethernet driver requires optimal implementation due to high-speed nature by reduced number of access (read/write) to the volatile register for the below APIs |
| Enhancement | [Linker files]: Unified Linker file design across different toolchains | [8375] Linker files updated for GCC and PPU Metaware in BaseframeworkTC4Dx |
| Enhancement | Request for Enhanced CRC32 Interface Implementation in ILLD for TC489 | [8377] Enhanced CRC32 Interface Implementation in ILLD to :<br>* Support data length up to 32-bit range<br>* Configurable endianness handling via compile-time switch<br>* Custom seed value initialization<br>* Multi-compiler compatibility |

| | | |
|---|---|---|
| Enhancement | solve access data no alignment by using memcpy api to copy data to destinationAddress IfxCan_Node_writeTxBufData,IfxCan_Node_writeData | [8521] 32-bit data copy replaced with bytewise data copy to prevent unaligned address access |
| Enhancement | TC4x Enhance the CAN loopback mode for LOUT feature | [8662] Support for LOUT feature has been added, similar to Loopback mode |
| Enhancement | Add a dummy trap handler to the TASKING PPU project. | [8738] Added default handler for tasking ppu |
| Enhancement | TASKING PPU Linker Scripts improvements | [8739] Added default handler and linker improvement for tasking ppu |
| Enhancement | Add AllowAccess to all template projects | [8740] Template project configuration updates done for all PPU devices |
| Enhancement | [TC4xx][HSPHY]: IfxHsphy_Geth_setupRgmiiOutputPins API configures the TX RGMII pins with the wrong speed grade | [8781] Speed grade updated for RGMII output pin configuration |
| Enhancement | Add IfxPpucCore_configureAccessToPpuVccm | [8789] Added API IfxPpucCore_configureAccessToPpuVccm() and IfxPpucStu_configureAccessToPpucStuDmi() |
| Enhancement | [TC4xx] I2C: DMA Support Request | [8808] APIs to support FIFO configuration for use with DMA |
| Enhancement | TC4x: 10 MHz XTAL clock support shall be removed for external customers | [8813] Removed the 10MHz XTAL support for external iLLD |
| Enhancement | iLLD for the CDSP GCC Compiler | [8853]Added support for CDSP illds for GCC compiler |
| Enhancement | allowAccess() function shall be called in Ifx_Cfg_Ssw.c::Ifx_Ssw_AP_Init function | [8963] The "allowAccess" function has been added for the TC4xx device and is now being called from the Ifx_Ssw_AP_Init function in Ifx_Cfg_Ssw.c file |
| Enhancement | Collect the files from the _Impl folder according to the device | [9020] Collected the config files in _Impl folder into subfolders according to the device |
| Enhancement | [TC4x][CPU] Unable to sync only specific cores in multi-core environment | [9043] The implementation of the IfxCpu_waitEvent doesn't not allow to synchronize only few specific cores (e..g. Core 2 versus Core 3) |
| Enhancement | [TC4x][STM] iLLD doesn't allow to init STM of another core | [9045] Updated IfxStm_initCompare Api to get index from passed cpu parameter |
| Enhancement | Consolidate IfxAdc_XXX_YYYSrvReq type which has same elements | [9046] Updated a common type structure IfxAdc_SrvReq in IfxAdc.h and utilized the same in different ADC files |

| | | |
|---|---|---|
| Enhancement | IfxCpu_configureAccessToCpus give access to CPUCS creating issue with AURIX VP | [9047] Conditional compilation added for IfxApApu_init of CPUCS in IfxCpu_configureAccessToCpus |
| Enhancement | IfxStm_configureAccessToStms give access to CPUCS creating issue with AURIX VP | [9048] Conditional compilation added for IfxApApu_init of CPUCS in IfxStm_configureAccessToStms |
| Enhancement | TC4x EGTM: TIM Driver updates needed for adding LUT to input source | [9152] Added LUT feature to Egtm Tim |
| Enhancement | QSPI: Provide APIs to get the QSPI error information | [9161] Added API to get the error status |
| Enhancement | QSPI: APIs to get and clear transmit/receive interrupt flags | [9163] Separate APIs ptovided to read and clear the interrupt flags |
| Enhancement | Update API IfxCan_Can_initNode() to initailize timeStampType of IfxCan_Can_Node | [9344] APIs have been updated to handle the timestamp value correctly |

## 3.2 Issues fixed in release V2.4.1

This is the first public release for the product, hence no issues are listed.

# 4 Known Issues

The known issues are listed in the below table

**Table 5**

| Key | Summary | Impact | Workaround |
|---|---|---|---|
| 0000061474-8763 | TMADC Issue's with group functionality and readings from result registers | If more than 7 channels are configured in a group and software buffer feature is used then system throws an exception since it access a reserved area | None |

# 5 Limitations and deviations

This chapter describes the limitations and deviations due to software/hardware design constraints.

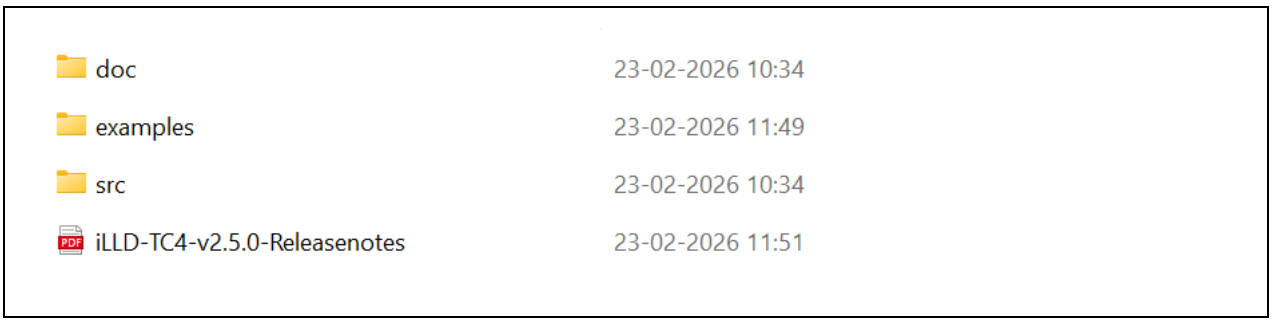
- None

# 6 Installation

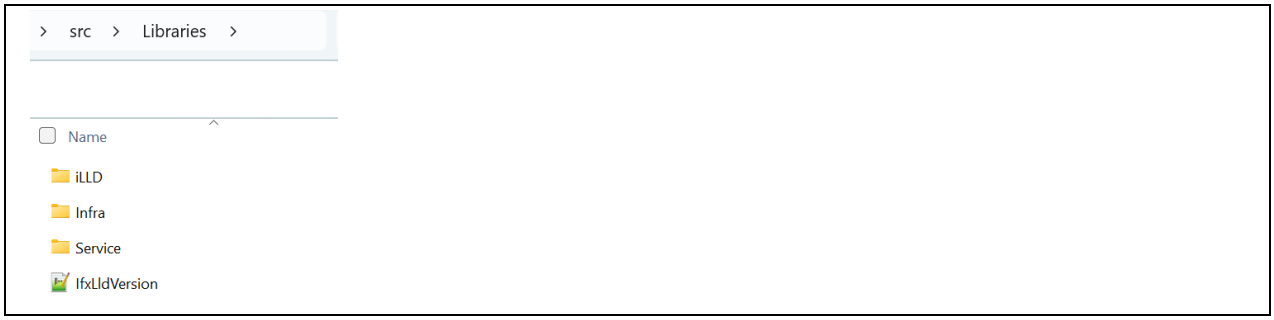
The iLLD software release package is an archive, and it may be extracted to any drive of the computer. There is no specific directory structure needed.

## 6.1 Package usage

- Copy the iLLD-TC4-<version>.zip file into local drive.
- Package contains the following items.
  - doc : Contains the user manual for the drivers
  - examples : Contains the template project for TC4Dx devices
  - src : Contains Source Code for the TC4Dx drivers
  - iLLD-TC4-<version>-Releasenotes : Aurix3G iLLD Release Note




  - src/Libraries/
    - iLLD : Directory contains the drivers for peripherals of different architecture
    - Infra : Directory contains sfr's, startup software and platform files
    - Service : Directory contains service layer libraries




  - src/Libraries/iLLD/TC4xx/
    - ArcEv : Directory contains drivers for ArcEv
    - Scr : Directory contains drivers for Scr
    - Tricore : Directory contains drivers for Tricore
    - CpuGeneric : Directory contains drivers common for both Tricore and ArcEv

| | |
|---|---|
| ArcEV | 04-07-2023 16:14 |
| CpuGeneric | 04-07-2023 16:14 |
| Scr | 04-07-2023 16:14 |
| Tricore | 16-11-2023 18:14 |

- The iLLD package contains drivers intended for multiple targets(Scr, Tricore, ArcEv). Template projects listed below are provided as part of package.

| Target | Template |
|---|---|
| Tricore | BaseFramework_TC4Dx<br>BaseFramework _TC4Dx_Scr<br>BaseFramework_TC4Dx_Ppu<br>HypervisorDemo_TC4Dx |

### Guidelines for SCR Baseframework

Within the SCR package, the i51 core drivers are exclusively compiled using the Tasking compiler tool. The Tricore, on the other hand, is compiled using the supported compiler tool that is currently available.

Once the compilation process is completed without errors, the next step involves flashing the Tricore elf. The Tricore will load a C array into the XRAM memory. This array contains all the necessary information required for the SCR boot-up process.

### Guidelines for Ppu Baseframework

The PPU drivers are compiled using the MetaWare Compiler. It is essential to verify the presence of predefined libraries for the TC4xx derivatives. If these libraries are included in the current version of MetaWare, you may proceed with the compilation process.

However, if the predefined libraries for the TC4xx derivatives are not available in the installed version of MetaWare on your system, please follow the steps below to install them:

1. Initiate the command prompt by pressing the combination of the Windows key + R.
2. Input the following command: *cmd /V:ON /k C:<MetaWare-path>setup.bat*. This command will activate the Python environment that is bundled with MetaWare.

To download the predefined libraries of a specific derivative, use the following command:

*build -ca -sc < derivative>_system -e BUILD_CONFIG=< derivative>_PPU -e BUILD_TYPE=debug*

For instance, for the 4Dx derivative, the command would be:

*build -ca -sc tc4Dx_system -e BUILD_CONFIG=TC4D_PPU -e BUILD_TYPE=debug*

This command will initiate the download of the predefined libraries for the specified derivative.

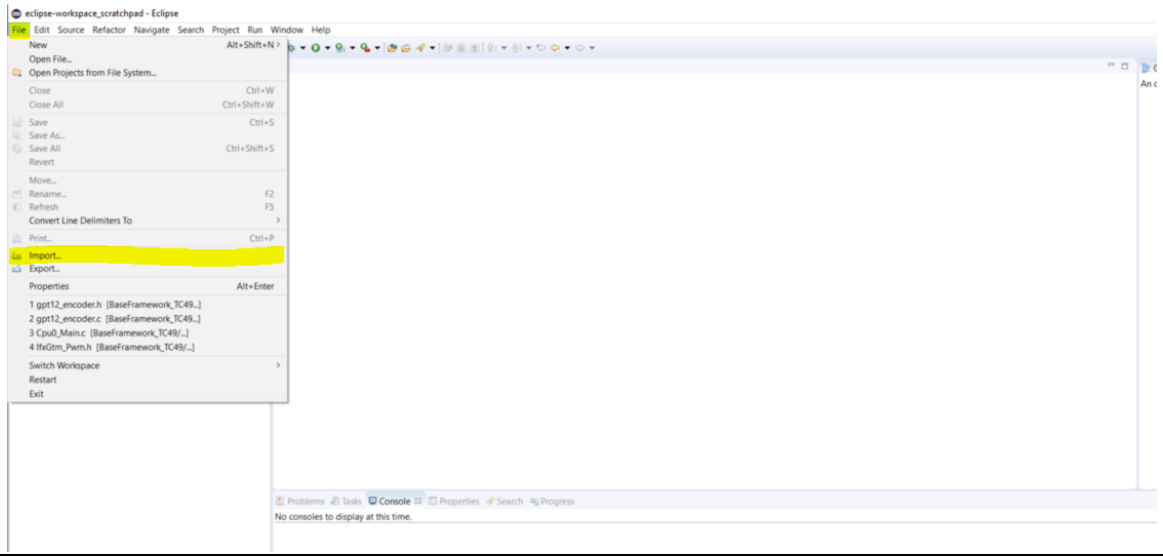
## 6.2 Importing Baseframework to Bifaces

- Import the template project from the package installation directory

  *C:\Infineon\iLLD-TC4-<version>/examples/<project>*

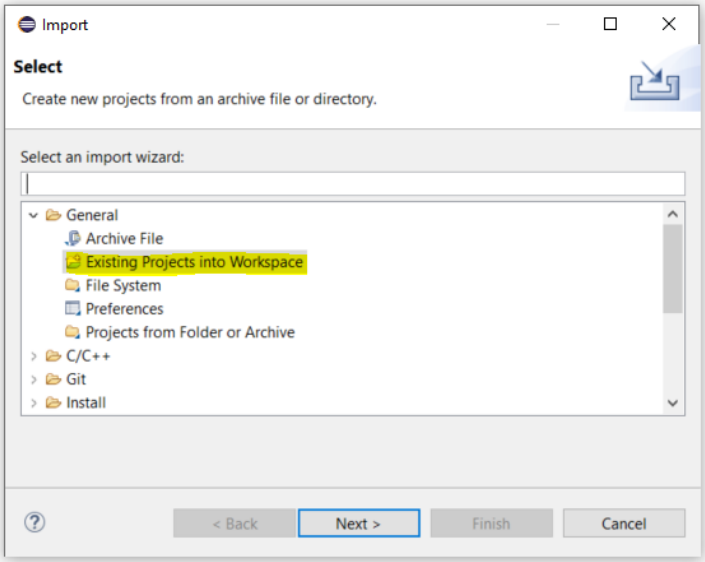
- Open Bifaces. Click on File --> Import

.

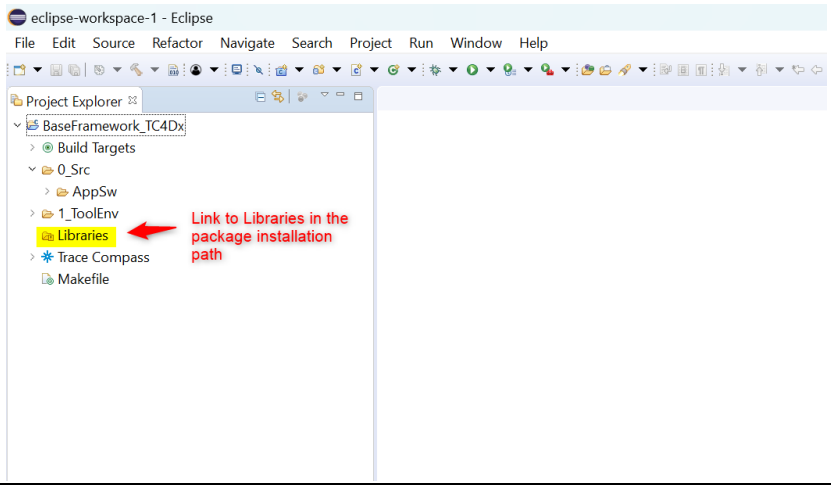



- Select existing project into workspace and Click Next.

.



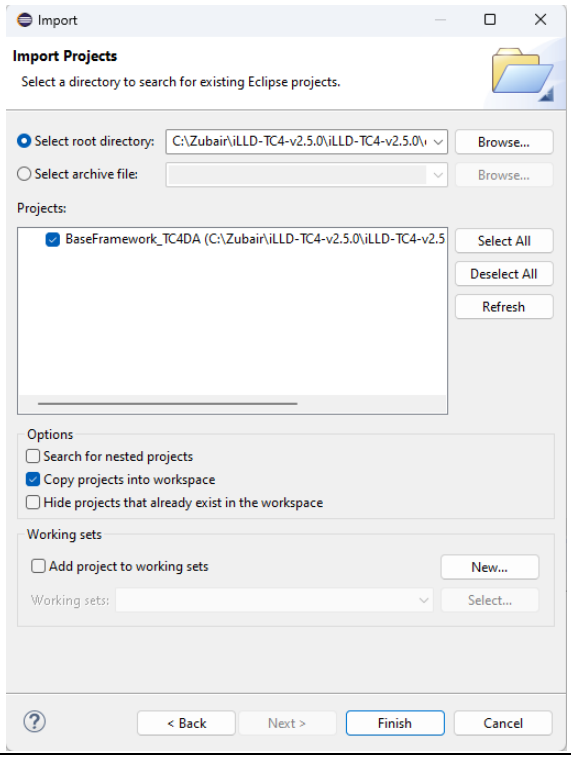
- Click on Browse and select the root directory where the Baseframework is present and Click ok.




*Note: The libraries are not included in the template projects, with the exception of the Hypervisor Baseframework. Instead, the libraries are located in the package installation path and are linked to the template project. This means that after importing the template project into BIFACES, users can compile the project as usual, and the required libraries will be automatically linked..*

- If you want to copy the Baseframework_TC4Dx into workspace, check the box copy project into workspace. Click Finish and the project will appear in Project Browser.

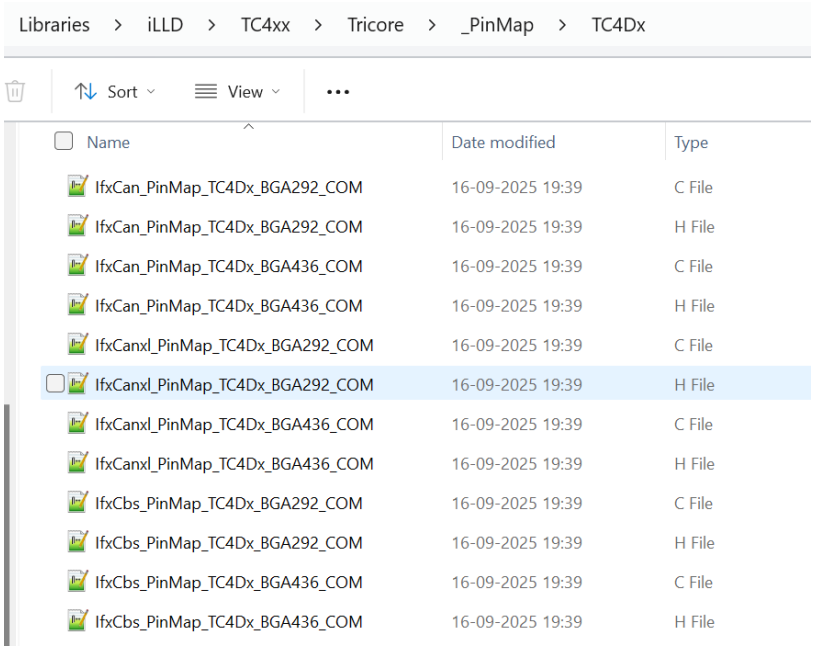
*Note: The libraries should also be copied to workspace along with the Baseframework.*
.




*Note: The iLLD package contains drivers intended for multiple targets (Tricore, SCR, ArcEv). To compile the drivers, please use the BaseFramework provided as reference, or ensure that drivers intended for SCR target are not built for Tricore.*

## 6.3 Selection of Pin package

- PinMap directory contains all the pinmaps for all the packages I.e. there are no separate folders for different packages.




- To select between the different packages, follow the below steps.
    - Packages can be selected with the help of the macros mentioned below, these should be defined in Ifx_Cfg.h
    - For example, in TC4Dx there are two packages available, so define macros as mentioned below
    - To use the BGA292_COM package, set the value of macro IFX_PIN_PACKAGE_BGA292_COM to 1
      #define IFX_PIN_PACKAGE_BGA292_COM 1
    - To use the BGA436_COM package, set the value of macro IFX_PIN_PACKAGE_BGA436_COM to 1
      #define IFX_PIN_PACKAGE_BGA436_COM 1

*Note: Only one macro should be defined in Ifx_cfg.h as explained in below screenshot.*

**File: Ifx_Cfg.h**

```
/* Change Package by defining macro */
#define IFX_PIN_PACKAGE_BGA292_COM 1
// #define IFX_PIN_PACKAGE_BGA436_COM 1
```

## 6.4 Compiler Selection to build iLLD

- Multiple compiler toolchains (mentioned in Table Tool information) are supported to build Tricore iLLD's
- Compiler selection is handled in Config.xml present in '**1_ToolEnv\0_Build\1_Config\Config.xml**'

.

```
<!-- Architecture congfigurations -->
<architecture name="Tricore" primaryToolchain= "Gnuc">
    <!-- Target congfigurations -->
    <target name="Tc" enable="true">
        <output type="elf" enable="true" fileName="$(PROJ_NAME)_Tc.elf" tool="Ld"/>
        <output type="hex" enable="true" fileName="$(PROJ_NAME)_Tc.hex" tool="Hex">
            <input siblingTypes= "elf" useObjects="false" />
        </output>
        <output type="srec" enable="false" fileName="$(PROJ_NAME)_Tc.srec" tool="Srec">
            <input siblingTypes= "elf" useObjects="false" />
        </output>
        <output type="lib" enable="false" fileName="$(PROJ_NAME)_Tc.a" tool="Ar"></output>
    </target>
```

## 6.5 Compiler Path to build iLLD

- Compile path is specified in '1_ToolEnv\0_Build\1_Config\Config_Tricore_{compiler}\Config_{compiler}.mk'

```
1_ToolEnv
  0_Build
    1_Config
      Config_ArcEV_Metaware
      Config_Tricore_Ghs
      Config_Tricore_Gnuc
        Config_Gnuc.mk
        Lcf_Gnuc_Tricore_Tc.lsl
      Config_Tricore_Hightec
      Config_Tricore_Tasking
    BifacesConfig.xsd
    Config.mk
    Config.xml
```

```
16 # MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE APPLY TO THIS SOFTWARE.#
17 # INFINEON SHALL NOT, IN ANY CIRCUMSTANCES, BE LIABLE FOR SPECIAL, INCIDENTAL,#
18 # OR CONSEQUENTIAL DAMAGES, FOR ANY REASON WHATSOEVER.                         #
19 #                                                                              #
20 ################################################################################
21
22 B_GNUC_TRICORE_PATH:= C:\Tools\HIGHTEC\gnutri_2020.03-b3513f1-win64
23
24
25 B_GNUC_TRICORE_CC_OPTIONS= -march=tc18 -mcpu=tc4xxx -g -O2 -fno-common -fstrict-volatile-bitfields \
26                            -ffunction-sections -fdata-sections -Wall -std=c11
27
28
29 B_GNUC_TRICORE_ASM_OPTIONS= $(GNUC_TC_CC_OPTIONS)
```

# 7 Support packages

No Support packages  are provided in this release

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